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Small-sided games as an integrated training strategy: effects on aerobic endurance and passing performance in adolescent soccer players – a randomized controlled trial

Sinan Seyhan[1*], Furkan Can Gündüz[1], Görkem Açar[2], Fatma Gözlükaya Girginer[3], Çağlar Soylu[4*], Yunus Emre Aslan[5] and Muhammed Fatih Bilici[6]

Abstract

Background Small-sided games (SSG) are increasingly used in youth soccer training because they simultaneously expose players to physiological, technical, and perceptual–cognitive demands similar to competitive match play. However, evidence regarding their combined effects on aerobic endurance and passing performance in adolescent players remains limited. Therefore, this study aimed to examine the effects of an eight-week SSG training program on aerobic endurance and LSPT performance in young male soccer players.

Methods Eighteen licensed youth male soccer players (mean age: 17 ± 0.4 years) were randomly allocated to an experimental SSG group ($n = 9$) or a control group ($n = 9$). The experimental group performed SSG-based training sessions three times per week for eight weeks, using formats that varied in pitch dimensions, player numbers, and tactical constraints to elicit high-intensity and decision-making stimuli. The control group continued their regular technical–tactical training without SSG exposure. Aerobic endurance and LSPT performance were assessed before and after the intervention using the Yo-Yo Intermittent Recovery Test Level 1 (Yo-Yo IR1) and the Loughborough Soccer Passing Test (LSPT). Data were analyzed using a mixed two-way ANOVA (group × time), and statistical significance was set at $p < 0.05$.

Results Significant group × time interactions were observed for both performance outcomes. The SSG group demonstrated substantial improvements in LSPT performance ($F(1,16) = 3844.00$, $p < 0.001$, $\eta^2 p = 0.996$) and Yo-Yo IR1 distance ($F(1,16) = 1648.60$, $p < 0.001$, $\eta^2 p = 0.990$), whereas the control group showed only trivial and non-significant changes. These results indicate that SSG training effectively associated with improvements both aerobic capacity and technical passing ability.

Conclusions An eight-week SSG training program was associated with improvements in aerobic endurance and LSPT performance in youth soccer players. SSG provide an ecologically valid and time-efficient training method by

*Correspondence:
Sinan Seyhan
sinan.seyhan@cbu.edu.tr
Çağlar Soylu
caglar.soylu@sbu.edu.tr

Full list of author information is available at the end of the article

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integrating metabolic load with technical and decision-making demands. Incorporating structured SSG formats into periodized training programs may support performance development in youth soccer.

Trial registration ClinicalTrials.gov, NCT07755358. Registered on 5 August 2026.

Keywords Small-sided games, Aerobic endurance, Passing accuracy, Youth soccer, Performance adaptation

Introduction

In recent years, football training methodologies have attracted growing interest within both performance science and training science research. Owing to its high-intensity and intermittent nature, football requires the concurrent engagement of aerobic and anaerobic energy systems. During a competitive match, players typically cover approximately 10–13 km while performing numerous high-intensity actions such as sprints, accelerations, decelerations, and directional changes [1–3]. These substantial physiological demands necessitate not only well-developed aerobic capacity but also advanced technical skills, tactical awareness, and cognitive processing abilities to sustain performance throughout match play [1, 2].

Within this context, small-sided games (SSG) have emerged as a widely adopted and scientifically supported training method for youth soccer development. By systematically manipulating pitch dimensions, player numbers, and task constraints, SSG are designed to reproduce match-specific physiological loads while simultaneously promoting technical and tactical behaviors. Previous research has demonstrated that 5v5 SSG formats significantly enhance maximal oxygen uptake (VO_2max) in young soccer players [4], while eight-week interventions using 4v4 plus goalkeeper formats have been shown to produce meaningful improvements in aerobic capacity and agility performance [5, 6]. In addition to their physiological benefits, SSG increase ball-contact frequency and passing opportunities, thereby facilitating faster decision-making processes and improvements in passing-related technical performance [7, 8].

In adolescent soccer players, SSG-based training may promote soccer-specific physiological, technical–tactical, and perceptual–cognitive adaptations. Repeated exposure to accelerations, decelerations, directional changes, and intermittent high-intensity efforts may contribute to improvements in aerobic endurance and recovery capacity. In addition, smaller playing areas and more frequent ball involvement may increase passing actions, ball control, decision-making speed, tactical awareness, and overall game intelligence [9].

Despite this growing body of evidence, much of the existing literature has predominantly focused on the physiological responses and aerobic adaptations induced by SSG-based training. In contrast, technical performance indicators—particularly passing accuracy—have received comparatively limited attention or have been assessed only as secondary outcomes [10, 11]. As a result, there remains a need for controlled experimental studies that concurrently examine both physiological and technical outcomes to provide a more integrated understanding of SSG-induced adaptations in youth football.

Therefore, the purpose of the present study was to investigate the effects of an SSG-based training program on aerobic endurance and passing accuracy in young soccer players. Aerobic endurance and passing performance, the study aims to contribute to the performance science literature and to inform evidence-based training practices for the holistic development of youth soccer players.

Materials and methods

Research design

This study was conducted using a pre-test–post-test control group experimental design. The study examined the effects of SSG training on aerobic endurance and passing accuracy in male soccer players aged 17.

The ethical suitability of the study was evaluated and approved by the Health Sciences Ethics Committee of Manisa Celal Bayar University. Ethical approval was obtained from the relevant committee with the decision dated July 17, 2025, and numbered 20.478.4.

This study was a two-arm, parallel-group randomized controlled trial. The design, conduct, and reporting of the trial adhere to the CONSORT 2010 (Consolidated Standards of Reporting Trials) statement and to the CONSORT extension for randomized trials of non-pharmacological treatments.

Power analysis

The sample size of the study was determined using G*Power 3.1.9.7 software. A power analysis was performed based on aerobic endurance within the scope of the "ANOVA: repeated measures, within–between interaction" analysis. A priori power analysis was conducted using G*Power 3.1.9.7 for a repeated-measures ANOVA with a within–between interaction. The analysis included two groups and two measurement points, with an alpha level of 0.05 and statistical power of 0.80. Based on an expected medium-to-large effect size, the required total sample size was estimated as 18 participants [5].

Study group

Eighteen male soccer players aged 17 years, with at least five years of soccer experience, playing in licensed

amateur soccer clubs voluntarily participated in the study. Participants were allocated to the experimental ($n = 9$) or control ($n = 9$) group using a stratified randomization procedure designed to ensure baseline comparability between groups. Stratification was performed on the basis of baseline anthropometric characteristics (height, body mass, and body mass index) and playing position, so that defenders, midfielders, forwards, and goalkeepers were represented in comparable proportions in each group. Within each stratum, participants were assigned to groups using a computer-generated random number sequence produced in Microsoft Excel (RAND function; Microsoft Corp., Redmond, WA, USA) by a researcher who was not involved in the delivery of the training intervention or in the administration of the performance tests. Group allocation was concealed until all baseline measurements had been completed. Due to the nature of the intervention, neither participants nor the coaches delivering the sessions could be blinded to group assignment; however, the researchers who administered and scored the Yo-Yo IR1 and LSPT tests were blinded to group allocation at both pre- and post-intervention assessments.

Certain criteria were considered in determining which football players would participate in the study. Inclusion criteria included participants being 17 years of age, having at least five years of football experience, regularly attending training sessions, not having suffered a serious injury in the last six months, and not having any chronic conditions. Conversely, the exclusion criteria included experiencing a serious injury, missing more than two training sessions, voluntarily requesting to withdraw from the study, or failing to participate in the test measurements.

Sociodemographic characteristics

Demographic and anthropometric characteristics, including age, height, body mass, body mass index (BMI), playing experience, dominant foot, and playing position, were recorded prior to the intervention (Table 1). Participants were between 17 and 18 years of age and had a minimum of five years of organized football experience (range: 5–10 years), indicating a relatively homogeneous youth football sample. The players represented a range of on-field positions, including defenders, midfielders, forwards, and goalkeepers, and included both right- and left-foot-dominant athletes. All participants were actively training and competing at the time of data collection and reported no musculoskeletal injuries or neurological conditions that could have influenced performance.

Data collection instruments and tests

Anthropometric measurements

Anthropometric assessments included body height, body mass, and BMI. Body height was measured to the nearest 0.1 cm using a stadiometer, with participants standing barefoot in an upright position according to the Frankfurt plane. Body mass was recorded to the nearest 0.1 kg using a calibrated mechanical scale, with participants wearing minimal clothing. BMI was calculated using the standard formula: body mass (kg) divided by height squared (m^2) [12].

Table 1 Demographic and anthropometric characteristics of the participants

Height (cm)	Body Mass (kg)	BMI (kg/m^2)	Age (years)	Dominant Foot	Years of Football Experience	Playing Position
184	70	20.7	17	Right	8	Center Back
186	82	23.7	17	Right	9	Goalkeeper
180	76	23.5	17	Right	7	Right Back
181	64	19.5	18	Left	6	Left Winger
177	65	20.8	18	Right	7	Central Midfielder
179	75	23.4	17	Right	8	Defensive Midfielder
176	67	21.6	17	Right	6	Right Winger
176	70	22.6	18	Left	7	Left Back
177	69	22.0	18	Right	5	Attacking Midfielder
187	79	22.6	17	Right	9	Center Back
177	63	20.1	17	Left	6	Left Winger
181	63	19.2	17	Right	5	Forward
176	64	20.7	17	Right	7	Right Winger
180	67	20.7	18	Right	8	Central Midfielder
175	65	21.2	18	Left	9	Left Back
180	67	20.7	17	Right	6	Center Back
179	65	20.3	17	Right	10	Forward
183	73	21.8	18	Right	7	Forward
Mean	179.67	69.11	21.39	17.39	–	7.22
SD	3.53	5.71	1.35	0.50	–	1.44

Yo-Yo intermittent recovery test level 1 (Yo-Yo IR1)

Aerobic endurance capacity was assessed using the Yo-Yo Intermittent Recovery Test Level 1 (Yo-Yo IR1). The test was conducted on a 2 × 20 m shuttle course with progressively increasing running speeds interspersed with 10 s of active recovery. Participants were required to maintain pace with audio signals, and the test was terminated when a participant failed to reach the finish line in time on two consecutive occasions. Total distance covered (m) was recorded as the performance outcome [13].

Loughborough soccer passing test (LSPT)

LSPT performance was evaluated using the Loughborough Soccer Passing Test (LSPT). Participants were instructed to complete a standardized sequence of 16 passes directed toward four target panels, following randomized color-coded commands. Time penalties were added for execution errors, including inaccurate passes, incorrect target selection, or ball control mistakes. Overall performance time was calculated as the sum of the execution time and penalty time, in accordance with established protocols [14].

Small-sided games (SSG) training protocol

The experimental group participated in a structured SSG training program conducted over an eight-week period, with three training sessions per week. Game formats (2v2, 3v3, and 4v4), pitch dimensions, and task constraints were progressively manipulated across the intervention period to increase training demands. Each session included a standardized 15-min warm-up and a 10-min cool-down. According to previous literature, SSG performed under these conditions typically elicit heart rate responses corresponding to 85–95% of maximal heart rate and blood lactate concentrations ranging between 4 and 8 mmol·L^{-1} [15] (Fig. 1.).

Both the experimental and control groups continued their regular club-based training schedules throughout the intervention period. Routine club training was performed five days per week and consisted of approximately 90 min per session, including a standardized 15-minute warm-up. These sessions primarily focused on technical passing drills, positional play, tactical instruction, and general conditioning exercises, and did not include structured small-sided games.

Importantly, training exposure was equivalent between groups with respect to weekly training frequency and routine content. In addition to this standardized club training, the experimental group completed three supplementary training sessions per week that exclusively consisted of structured SSG. These additional SSG sessions lasted approximately 30 min and were conducted on separate occasions from regular training.

The SSG formats included 4v4, 3v3, and 2v2 configurations, implemented using a consistent 1:1 work-to-rest ratio. Pitch dimensions were systematically adjusted according to the number of players involved to maintain comparable relative playing areas, with dimensions set at 35 × 25 m for 4v4, 30 × 20 m for 3v3, and 25 × 15 m for 2v2 formats. Once per week, a 3v3 + 2 neutral players (jokers) variation was incorporated during the 4v4 sessions to increase passing options, ball circulation, and tactical complexity. This design ensured that both groups received equivalent exposure to routine football training, while providing the experimental group with an additional structured SSG stimulus. It should be noted, however, that because the experimental group performed three additional 30-minute SSG sessions per week, training type (SSG-based stimulus) and training volume (extra weekly exposure) cannot be fully disentangled in the present design (see Limitations).

Data collection procedure

All assessments were conducted on the same outdoor football pitch, using identical equipment and at consistent times of day to minimize environmental and circadian influences. Baseline (pre-test) measurements were performed one week prior to the commencement of the training intervention, while post-test measurements were collected at the end of the eight-week training period. All tests were administered under standardized conditions and supervised by the same research team to ensure consistency and reliability of data collection.



Fig. 1 Representative examples of the small-sided games (SSG) training protocol. **A** Change-of-direction and acceleration drills integrated into the SSG warm-up phase. **B** Linear and multi-directional running patterns performed within constrained pitch dimensions to increase movement variability and physiological load

Statistical analysis

All statistical analyses were performed using IBM SPSS Statistics software (version 26.0; IBM Corp., Armonk, NY, USA). Descriptive statistics were calculated for all variables and reported as mean ± standard deviation, along with minimum and maximum values. Data normality was assessed using the Shapiro–Wilk test in conjunction with skewness and kurtosis values.

To evaluate baseline homogeneity between the experimental and control groups, independent samples *t*-tests were applied. Within-group pre- to post-intervention comparisons were conducted using paired samples *t*-tests. To examine the effects of the training intervention, a two-way mixed-design analysis of variance (ANOVA) with one between-subjects factor (Group: experimental vs. control) and one within-subjects factor (Time: pre-test vs. post-test) was performed. The level of statistical significance was set at $p < 0.05$. Effect sizes were reported as partial eta squared ($\eta^2 p$) and interpreted as small (≥ 0.01), medium (≥ 0.06), or large (≥ 0.14) effects. Prior to analysis, all statistical assumptions, including normality, homogeneity of variances, and equality of covariance matrices, were verified. To complement significance testing and to provide information on the precision and practical magnitude of the observed effects, mean changes from pre- to post-intervention were calculated within each group, and the between-group difference in mean change (representing the Time × Group interaction effect) was reported as the primary treatment effect estimate. All mean change estimates are accompanied by 95% confidence intervals, derived from the pooled within-subjects error term of the two-way mixed ANOVA (df = 16) under the equal-variance assumption of the model. Percentage changes from baseline are also reported to facilitate comparison with previous literature.

Results

The results of this study are presented for the two primary outcome measures: LSPT performance and Yo-Yo IR1 distance. The primary outcome measures of the study were LSPT performance and Yo-Yo IR1 distance, which were analyzed separately in the subsequent statistical analyses.

Prior to the intervention, both experimental and control groups demonstrated statistically comparable baseline performance levels in LSPT and Yo-Yo IR1 assessments ($p > 0.05$), confirming homogeneity between groups. Normality analyses (Shapiro–Wilk tests) verified that all variables were normally distributed, allowing for the use of parametric procedures.

All 18 players who were assessed for eligibility met the inclusion criteria, were randomized (SSG group, $n = 9$; control group, $n = 9$), and received their allocated intervention. No participant was lost to follow-up, discontinued the intervention, or was excluded from the analysis; therefore, data from all 18 participants ($n = 9$ per group) were analysed in the groups to which they had been randomly allocated. Recruitment and the eight-week intervention were conducted between September and November 2025, and the trial ended when the pre-specified eight-week intervention period and post-testing had been completed in all participants. No injuries, adverse events, or other unintended effects attributable to the SSG intervention or to the performance testing were observed or reported in either group. The flow of participants through the trial is summarized in Fig. 2.

Following the eight-week intervention, marked improvements were observed in the experimental group across both performance domains. LSPT results indicated a significant reduction in total completion time, reflecting enhanced passing precision and decision-making speed. Similarly, Yo-Yo IR1 findings revealed a substantial increase in total distance covered, denoting improved aerobic endurance and recovery efficiency. Two-way mixed ANOVA analyses further demonstrated highly significant Time × Group interactions in both variables ($p < 0.001$, $\eta^2 p > 0.990$), confirming that the SSG training protocol produced superior gains compared with traditional training routines.

Descriptive statistics, within-group mean changes, and 95% confidence intervals for pre- and post-test LSPT scores in the experimental and control groups are presented in Table 2. As shown in Table 2, the mean LSPT completion time of the experimental group decreased markedly from 40.99 ± 0.33 s at the pre-test to 36.90 ± 0.27 s at the post-test, corresponding to a within-group mean change of − 4.09 s (95% CI: −4.18, − 4.00), or a − 9.98% reduction in completion time. This result indicates a substantial improvement in technical passing performance following the eight-week SSG training program. In contrast, the control group exhibited only a minor reduction from 40.13 ± 0.82 s to 39.83 ± 0.94 s over the same period, with a within-group mean change of − 0.30 s (95% CI: −0.39, − 0.21) and a percentage change of − 0.75%. The non-overlapping 95% confidence intervals between the two groups further support the conclusion that the magnitude of improvement in the experimental group substantially exceeded that observed in the control group.

This descriptive trend suggests that participation in the SSG-based training protocol led to a meaningful enhancement in both passing accuracy and execution speed compared with traditional training routines. These findings were further supported by inferential analyses, which confirmed the statistically significant effects of time and group × time interaction on LSPT performance ($p < 0.001$).

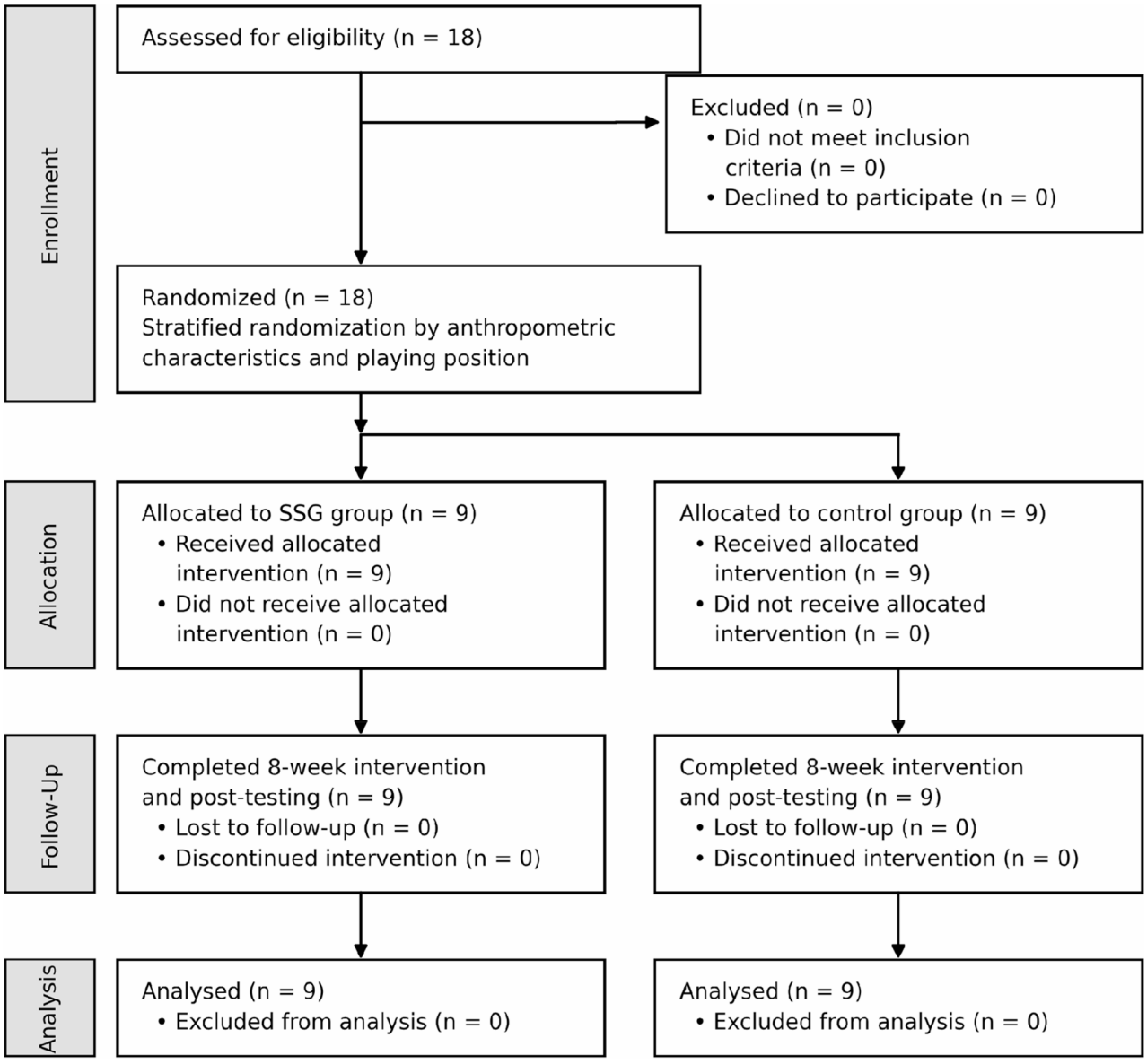


Fig. 2 CONSORT flow diagram showing the progress of participants through each stage of the trial

Table 2 Descriptive statistics, within-group mean changes, and 95% confidence intervals for loughborough soccer passing test (LSPT) completion time

Group	*n*	Pre-test Mean ± SD (min–max)	Post-test Mean ± SD (min–max)	Within-group mean change (95% CI)	% change
Experimental	9	40.99 ± 0.33 (40.5–41.5)	36.90 ± 0.27 (36.5–37.3)	−4.09 (−4.18, −4.00)	−9.98%
Control	9	40.13 ± 0.82 (38.5–41.1)	39.83 ± 0.94 (37.9–41.0)	−0.30 (−0.39, −0.21)	−0.75%

Negative values indicate a reduction in total completion time (i.e., improved passing performance). 95% confidence intervals (CIs) for the within-group mean change were derived from the pooled within-subjects error term of the two-way mixed ANOVA (df = 16)

SD Standard deviation

As shown in Table 3, both groups exhibited skewness and kurtosis values within the acceptable range of ± 1, indicating a near-normal distribution. The Shapiro–Wilk test further confirmed that the LSPT data did not significantly deviate from normality for either the experimental group (SW = 0.945, $p = 0.358$) or the control group (SW = 0.933, $p = 0.217$).

The Shapiro–Wilk test indicated that the distribution of LSPT scores did not significantly deviate from normality in either group ($p > 0.05$), supporting the assumption of normality and justifying the use of parametric analyses,

Table 3 Normality statistics for loughborough passing test (LSPT) scores

Group	*n*	Skewness	Kurtosis	Shapiro–Wilk	*p* value
Experimental	9	0.215	0.450	0.945	0.358
Control	9	−0.854	0.602	0.933	0.217

Normality was assessed using skewness and kurtosis values together with the Shapiro–Wilk test. $p > 0.05$ indicates that the data do not significantly deviate from a normal distribution

including the independent-samples t test and two-way mixed ANOVA, in subsequent analyses.

As presented in Table 4, there was no statistically significant difference in baseline LSPT completion time between the experimental group (40.99 ± 0.33 s) and the control group (40.13 ± 0.82 s), $t(16) = 1.046$, $p = 0.311$. The mean difference between groups was 0.86 s (95% CI: −0.88 to 2.59), with the confidence interval including zero, further indicating that the groups were comparable at baseline in terms of technical passing performance prior to the intervention.

As shown in Table 5, the two-way mixed ANOVA revealed significant effects for time, group, and the Time × Group interaction. The significant main effect of time indicates that LSPT completion time decreased from pre- to post-test across the entire sample. In addition, the significant main effect of group suggests overall differences in LSPT performance between the experimental and control groups across both measurement points.

More importantly, the significant Time × Group interaction demonstrates that the magnitude of change over time differed substantially between groups. Specifically, the experimental group exhibited a markedly greater reduction in LSPT completion time following the SSG intervention, whereas the control group showed only a minor improvement.

Crucially, a highly significant *Time* × *Group* interaction was detected ($F(1,16) = 3844.000$, $p < 0.001$, $\eta^2 p = 0.996$), demonstrating that the magnitude of change in LSPT performance over time differed markedly between groups (Table 5). Post-intervention comparisons indicated that the experimental group exhibited a pronounced reduction in LSPT completion time following the SSG training program, whereas the control group showed only marginal improvement.

The distribution of individual responses and between-group differences in pre- to post-intervention changes are illustrated in Fig. 3, which visually supports the strong interaction effect observed in the mixed ANOVA. Collectively, these findings indicate that SSG-based training was associated with substantially greater improvements in passing execution speed compared with regular training in this cohort of young soccer players.

As presented in Table 6, the experimental group demonstrated a marked improvement in aerobic endurance following the intervention. The mean distance covered during the Yo-Yo IR1 test increased from 1200.00 ± 136.93 m at pre-test to 1505.56 ± 146.72 m at post-test, corresponding to a mean improvement of 305.56 m (95% CI: 296.94 to 314.18) and a 25.46% increase.

By comparison, the control group exhibited only a modest improvement, with mean Yo-Yo IR1 distance increasing from 1220.00 ± 135.37 m at pre-test to 1292.22 ± 136.27 m at post-test. This corresponded to a mean increase of 72.22 m (95% CI: 63.60 to 80.84) and a 5.92% improvement over the intervention period.

These descriptive results indicate that participation in the SSG training program elicited pronounced improvements in high-intensity intermittent running capacity compared with traditional training routines. The magnitude of this improvement suggested that the intervention effectively enhanced players' aerobic endurance and recovery efficiency under game-like conditions.

Table 4 Independent samples t-test results for pre-test loughborough soccer passing test (LSPT) scores, including the 95% confidence interval for the baseline mean difference

Group	*n*	Mean	SD	Mean difference (95% CI)	t	df	*p* value
Experimental	9	40.989	0.326	+0.86 (−0.88, +2.59)	1.046	16	0.311
Control	9	40.133	0.817				

The baseline mean difference is reported as Experimental minus Control. The 95% CI includes zero, confirming baseline comparability between groups

SD Standard deviation

Table 5 Two-way mixed ANOVA results for the effect of the training program on loughborough passing test (LSPT) performance

Source	Sum of Squares	df	Mean Square	F	*p* value	Partial η^2
Between-subjects effects						
Group	9.714	1	9.714	11.349	0.004	0.415
Error	13.694	16	0.856			
Within-subjects effects						
Time	43.340	1	43.340	5,157.851	< 0.001	0.997
Time × Group	32.300	1	32.300	3,844.000	< 0.001	0.996
Error	0.134	16	0.008			

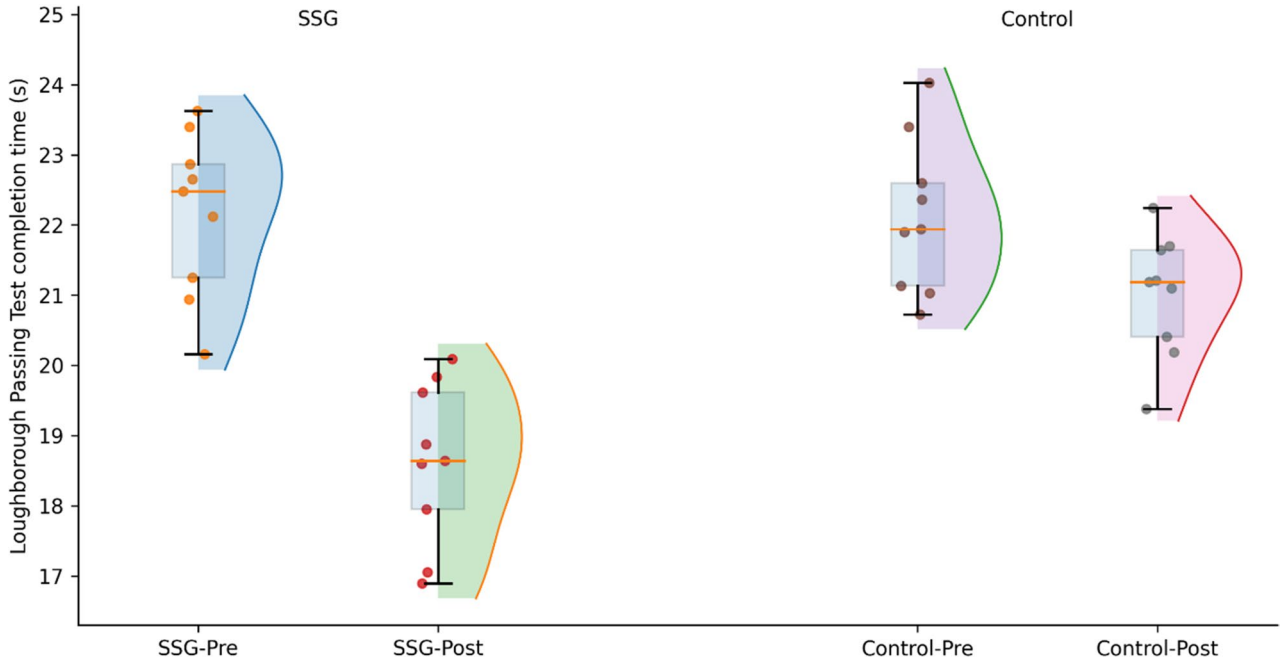


Fig. 3 Individual and group-level changes in passing performance following the eight-week intervention. Raincloud plot illustrating pre- and post-intervention changes in Loughborough Soccer Passing Test (LSPT) completion time in the small-sided games (SSG; $n = 9$) and control ($n = 9$) groups. The plot combines individual data points, boxplots (median and interquartile range), and kernel density distributions to display within- and between-group variability. A significant *Time* × *Group* interaction indicates a markedly greater reduction in LSPT completion time following the SSG intervention compared with the control condition

Table 6 Descriptive statistics, within-group mean changes, and 95% confidence intervals for Yo-Yo intermittent recovery test level 1 (Yo-Yo IR1) distance

Group	*n*	Pre-test Mean ± SD (min–max)	Post-test Mean ± SD (min–max)	Within-group mean change (95% CI)	% change
Experimental	9	1200.00 ± 136.93 (1000–1400)	1505.56 ± 146.72 (1300–1750)	+ 305.56 (+ 296.94, + 314.18)	+ 25.46%
Control	9	1220.00 ± 135.37 (1020–1410)	1292.22 ± 136.27 (1090–1490)	+ 72.22 (+ 63.60, + 80.84)	+ 5.92%

Positive values indicate an increase in distance covered (i.e., improved aerobic endurance). 95% confidence intervals (CIs) for the within-group mean change were derived from the pooled within-subjects error term of the two-way mixed ANOVA (df = 16)

SD Standard deviation

Table 7 Normality test results for Yo-Yo intermittent recovery test level 1 (Yo-Yo IR1) scores

Group	Time Point	*n*	Shapiro–Wilk	*p* value
Experimental	Pre-test	9	0.972	0.914
	Post-test	9	0.982	0.975
Control	Pre-test	9	0.964	0.839
	Post-test	9	0.971	0.905

Normality was assessed using the Shapiro–Wilk test. $p > 0.05$ indicates that the data do not significantly deviate from a normal distribution

As shown in Table 7, all *p*-values obtained from the Shapiro–Wilk tests exceeded the 0.05 significance threshold, indicating no significant deviation from normality across any measurement point. Both the experimental and control groups displayed symmetrical distributions for pre- and post-test scores, with SW values ranging from 0.964 to 0.982.

The Shapiro–Wilk test indicated that Yo-Yo IR1 scores were normally distributed in both groups ($p > 0.05$), confirming that the assumption of normality was met and supporting the use of parametric analyses, including the independent-samples t test and two-way mixed ANOVA.

As shown in Table 8, there was no statistically significant difference in baseline Yo-Yo IR1 performance between the experimental group (1200.00 ± 136.93 m) and the control group (1220.00 ± 135.37 m), $t(16) = -0.312$, $p = 0.759$. The mean difference between groups was − 20.00 m (95% CI: −155.9 to 115.9), and the confidence interval included zero, indicating that the groups were comparable in aerobic endurance performance prior to the intervention.

This result confirms that both groups started the intervention period with comparable aerobic capacities, ensuring that any subsequent improvements observed in the post-test phase could be attributed to the SSG training program rather than to pre-existing fitness differences.

A two-way mixed analysis of variance (ANOVA) was conducted to examine the effects of the eight-week SSG

Table 8 Independent samples t-test results for baseline Yo-Yo intermittent recovery test level 1 (Yo-Yo IR1) scores, including the 95% confidence interval for the baseline mean difference

Group	*n*	Mean (m)	SD	Mean difference (95% CI)	t	df	*p* value
Experimental	9	1200.00	136.93	−20.00 (− 155.9, + 115.9)	−0.312	16	0.759
Control	9	1220.00	135.37				

The baseline mean difference is reported as Experimental minus Control. The 95% CI includes zero, confirming baseline comparability between groups

SD Standard deviation

Table 9 Two-way mixed ANOVA results for Yo-Yo intermittent recovery test level 1 (Yo-Yo IR1) performance

Source	Sum of Squares	df	Mean Square	F	*p* value	Partial η^2
Between-subjects effects						
Group	84100.000	1	84100.000	2.184	0.159	0.120
Error	616188.889	16	38511.806			
Within-subjects effects						
Time	321111.111	1	321111.111	4321.495	< 0.001	0.996
Time × Group	122500.000	1	122500.000	1648.598	< 0.001	0.990
Error	1188.889	16	74.306			

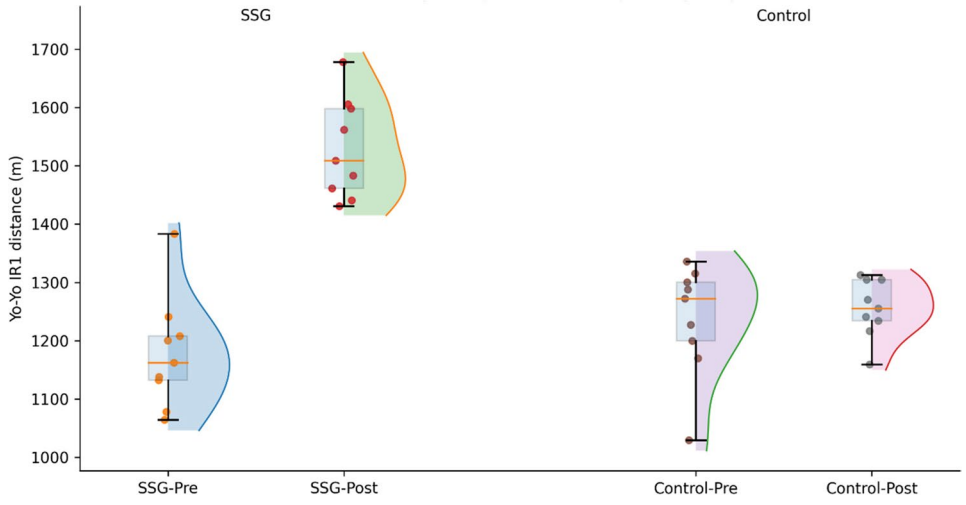


Fig. 4 Individual and group-level changes in aerobic endurance following the eight-week intervention. Raincloud plot illustrating pre- and post-intervention changes in Yo-Yo Intermittent Recovery Test Level 1 (Yo-Yo IR1) performance in the small-sided games (SSG; $n = 9$) and control ($n = 9$) groups. The plot combines individual data points, boxplots (median and interquartile range), and kernel density distributions to display within- and between-group variability. A significant *Time* × *Group* interaction indicates substantially greater improvements in Yo-Yo IR1 performance in the SSG group compared with the control group

training program on aerobic endurance as measured by the Yo-Yo IR1 test. The results are displayed in Table 9.

The two-way mixed ANOVA demonstrated a significant main effect of time on Yo-Yo IR1 performance ($F(1,16) = 4321.495$, $p < 0.001$, $\eta^2 p = 0.996$), indicating a substantial improvement in aerobic performance from pre- to post-intervention across all participants (Table 9). In contrast, the main effect of group was not statistically significant ($F(1,16) = 2.184$, $p = 0.159$, $\eta^2 p = 0.120$), suggesting that when pre- and post-test values were considered together, overall performance levels did not differ significantly between the experimental and control groups.

Importantly, a highly significant *Time* × *Group* interaction was observed ($F(1,16) = 1648.598$, $p < 0.001$, $\eta^2 p = 0.990$), indicating that the magnitude of improvement over time differed markedly between groups (Table 9). Post-intervention analyses revealed that the experimental group demonstrated a substantially greater increase in Yo-Yo IR1 distance following the small-sided games (SSG) training program, whereas the control group exhibited only a modest improvement.

The distribution of individual responses and between-group differences in pre- to post-intervention changes are illustrated in Fig. 4. Although the partial eta-squared value for the Time × Group interaction ($\eta^2 p = 0.990$) is numerically very large, this estimate must be interpreted with caution given the small sample size and the limited within-group variability of the present cohort, both of which are known to inflate partial eta-squared

Table 10 Between-group comparison of pre- to post-intervention mean changes in primary outcomes, representing the estimated treatment effect of the eight-week SSG intervention

Outcome	Experimental Δ (95% CI)	Control Δ (95% CI)	Between-group difference in Δ (95% CI)	F(1, 16)	*p*	η^2p
LSPT	−4.09 (−4.18, −4.00)	−0.30 (−0.39, −0.21)	−3.79 (−3.92, −3.66)	3844.00	< 0.001	0.996
Yo-Yo IR1	+305.56 (+296.94, +314.18)	+72.22 (+63.60, +80.84)	+233.34 (+221.16, +245.52)	1648.60	< 0.001	0.990

Δ = (post-test – pre-test). The between-group difference in Δ represents the estimated treatment effect of the SSG intervention relative to the control condition (i.e., the Time × Group interaction contrast). 95% confidence intervals were derived from the pooled within-subjects error term of the two-way mixed ANOVA (df = 16). F and p values correspond to the Time × Group interaction. Partial eta-squared (η^2p) values should be interpreted cautiously given the small sample size and limited within-group variability (see Discussion, Limitations)

LSPT Loughborough Soccer Passing Test, *Yo-Yo IR1* Yo-Yo Intermittent Recovery Test Level 1

(see Limitations). Nevertheless, the clear separation of post-intervention distributions and the consistent direction of individual responses, together with mean changes that exceeded established smallest-worthwhile-change and minimal-detectable-change thresholds, indicate that SSG-based training elicited practically meaningful gains in aerobic endurance compared with regular training in this cohort of young soccer players.

As presented in Table 10, the experimental group demonstrated substantially greater improvements than the control group in both primary outcomes following the eight-week SSG intervention.

For LSPT performance, the experimental group showed a mean reduction of 4.09 s (95% CI: −4.18 to −4.00), whereas the control group improved by only 0.30 s (95% CI: −0.39 to −0.21). The between-group difference in change scores was −3.79 s (95% CI: −3.92 to −3.66), indicating a significantly greater improvement in passing performance in the experimental group. The Time × Group interaction was statistically significant, $F(1, 16) = 3844.00$, $p < 0.001$, with an extremely large effect size ($\eta^2p = 0.996$).

Similarly, for Yo-Yo IR1 performance, the experimental group improved by 305.56 m (95% CI: 296.94 to 314.18), whereas the control group demonstrated an increase of 72.22 m (95% CI: 63.60 to 80.84). The between-group difference in mean change was 233.34 m (95% CI: 221.16 to 245.52), indicating a markedly greater enhancement in aerobic endurance in the experimental group. The Time × Group interaction was also statistically significant for Yo-Yo IR1, $F(1, 16) = 1648.60$, $p < 0.001$, with an extremely large effect size ($\eta^2p = 0.990$).

Discussion

The present study investigated the effects of an eight-week SSG training program on aerobic endurance and LSPT performance in young soccer players. The main findings demonstrated statistically significant improvements in both aerobic performance, assessed via the Yo-Yo IR1, and LSPT performance, measured using the LSPT. Beyond statistical significance, the magnitude of these changes suggests that the observed improvements are likely to be practically meaningful for youth soccer performance when interpreted within a match-related context. Nevertheless, the interpretation of technical outcomes requires careful consideration of the conceptual and ecological limitations inherent to the selected measurement tools.

The experimental group exhibited a mean improvement of approximately + 305 m in Yo-Yo IR1 performance. In youth football populations, the smallest worthwhile change (SWC) for intermittent endurance tests is commonly estimated as ~ 0.2 × between-subject standard deviation, corresponding to roughly 120–150 m in comparable cohorts [5, 13]. Accordingly, the observed improvement substantially exceeded the SWC threshold, indicating that the gain was not only statistically significant but also meaningful in performance terms.

Furthermore, the change magnitude clearly surpassed the minimal detectable change (MDC) reported for Yo-Yo IR1 in youth players, which ranges between 90 and 120 m, depending on testing conditions and population characteristics [13]. This supports the interpretation that the observed increase reflects a true physiological adaptation, rather than measurement error or day-to-day variability.

From a match-performance perspective, improvements of this magnitude are associated with enhanced high-intensity running capacity, improved recovery between repeated efforts, and greater tolerance to congested match schedules. Previous studies have demonstrated strong relationships between Yo-Yo IR1 performance and match-derived indicators such as total distance covered, high-speed running, and repeated-sprint involvement [16–18]. Consequently, the aerobic gains observed in the present study are likely to translate into improved match-running performance and sustained technical involvement, particularly during later phases of play in youth competitions characterized by frequent transitions.

The experimental group demonstrated a substantial reduction in LSPT completion time (≈ 4.1 s), exceeding commonly reported SWC values for time-based passing tests, which typically range between 1.0 and 1.5 s in youth footballers [14, 19]. Moreover, this improvement surpassed the reported MDC for LSPT-derived measures, suggesting that the observed change reflects a real improvement in task execution, rather than test–retest variability.

However, it is essential to acknowledge that the LSPT represents a controlled, closed-skill assessment conducted in a non-pressured, non-oppositional environment. As highlighted by Serpiello et al. the LSPT demonstrates limited criterion validity in elite youth football populations, particularly when the aim is to infer in-game passing performance [14]. Consequently, improvements in LSPT performance should not be interpreted as direct evidence of enhanced match-level passing accuracy, but rather as improvements in isolated technical execution speed and motor coordination.

This distinction is particularly relevant given that passing performance in competitive football is strongly influenced by contextual and perceptual–cognitive constraints, including opponent pressure, time availability, spatial uncertainty, and decision-making demands. The LSPT does not capture these critical dimensions of performance and therefore cannot fully reflect the ecological complexity of match play.

Despite these limitations, the LSPT improvements observed remain meaningful when interpreted as indicators of technical readiness rather than direct match performance. SSG inherently impose high perceptual and cognitive loads by increasing ball-contact frequency, reducing available space, and accelerating decision-making demands [7, 20]. These conditions are known to enhance perception–action coupling and motor efficiency, which may partially transfer to improved execution speed in controlled technical tasks such as the LSPT.

Previous studies have similarly reported that SSG-based interventions produce superior gains in short-passing execution compared with isolated drills or repeated-sprint training, even when assessed using decontextualized tests [19, 21]. Therefore, while the LSPT does not directly measure in-game passing success, the magnitude of improvement observed in the present study likely reflects underlying neuromotor and perceptual adaptations fostered by SSG exposure.

Nevertheless, the absence of match-derived technical indicators—such as successful passes under pressure, decision-making accuracy, or positional data—represents a clear limitation. As such, the present findings should be interpreted as evidence that SSG enhance foundational technical efficiency, rather than definitive proof of improved in-game passing performance.

Although some authors have suggested that repeated-sprint or interval-based protocols may yield superior aerobic adaptations [22], the present findings support the growing consensus that SSG can produce comparable aerobic gains while simultaneously enhancing technical efficiency. Zaharia et al. demonstrated that 4v4 + goalkeeper SSG improved VO_2max and agility more effectively than traditional endurance training [5], while Arslanoglu et al. reported additive benefits when SSG were combined with maximal aerobic-speed drills [23].

The dual-stimulus nature of SSG—simultaneously engaging metabolic, neuromuscular, perceptual, and cognitive systems—likely underpins these adaptations. By repeatedly exposing players to accelerations, decelerations, directional changes, and rapid information processing in representative environments, SSG reproduce key mechanical and cognitive demands of competition more effectively than linear conditioning drills [24].

Manipulating pitch size, player number, and rule constraints remains a critical factor in targeting specific adaptations. Larger relative areas per player (≥ 150 m^2) appear to accentuate aerobic and high-speed running demands, whereas smaller spaces (≤ 100 m^2) promote frequent technical actions and accelerated decision-making [18]. Coaches may therefore strategically sequence these formats to balance physical conditioning with technical and perceptual development.

Future research should extend these findings by incorporating match-derived performance metrics (e.g., GPS-based passing networks, successful passes under defensive pressure) to address the ecological limitations of closed-skill tests such as the LSPT. Additionally, integrating neurocognitive and biomechanical assessments may further clarify the mechanisms underlying SSG-induced improvements. Comparative analyses between SSG, hybrid game-profile-based training approaches [25–27], and technology-assisted monitoring systems could refine evidence-based training prescriptions for youth football development.

The findings of this study should be interpreted in light of several limitations. First, the relatively small sample size ($n = 18$) constrains the generalizability of the findings and warrants particular caution in the interpretation of the reported effect size estimates. The partial eta-squared values observed for the Time × Group interactions ($\eta^2 p > 0.99$) are exceptionally large and likely subject to upward bias. Partial eta-squared is mathematically sensitive to the magnitude of residual (error) variance: when within-group variability is low—as evident in the present cohort, where post-test standard deviations in the experimental group were small (LSPT: 0.27 s; Yo-Yo IR1: 146.72 m)—the error term becomes minimal and the proportion of variance attributable to the effect becomes disproportionately inflated. This bias is further amplified in small samples, in which variance components are estimated with reduced precision and partial eta-squared is known to be positively biased relative to its population value. Consequently, the η^2p values reported here should be regarded as sample-specific descriptors of the relative magnitude of within-sample variation rather than as unbiased population-level estimates, and they should be interpreted in conjunction with the raw mean

changes, individual response distributions, and the practical thresholds (SWC and MDC) discussed above, which provide more ecologically meaningful indicators of the training response. Future studies employing larger and more heterogeneous samples are needed to obtain more stable effect size estimates, ideally complemented by bias-corrected metrics (e.g., omega-squared, ω^2) or bootstrap-derived confidence intervals. Although the Yo-Yo Intermittent Recovery Test Level 1 is a valid indicator of intermittent aerobic capacity, internal load variables (e.g., heart rate or blood lactate) were not directly measured during the small-sided games sessions, restricting precise dose–response interpretation. A further consideration concerns the potential confounding of training type and training volume. Whereas both groups continued their routine club training of approximately five sessions of 90 min per week, the experimental group completed three additional 30-minute SSG sessions per week (≈90 min/week of supplementary exposure), resulting in a modestly higher total weekly training load (~20%) compared with the control group. Although these supplementary sessions were deliberately designed to isolate the SSG stimulus and were brief in duration, the present design does not allow training type (SSG-based stimulus) and training volume (additional weekly exposure) to be fully disentangled. Consequently, the between-group differences observed in aerobic endurance and passing performance cannot be attributed exclusively to the qualitative nature of the SSG stimulus, and a partial contribution from the increased total training exposure cannot be ruled out. Future research should adopt matched-volume designs, in which the control group performs an equivalent duration of alternative training (e.g., generic conditioning, drill-based technical work, or continuous running) in place of SSG, to more clearly isolate the specific effects of the SSG modality from those of additional training exposure. Technical performance was assessed using the Loughborough Passing Test, a closed-skill and non-oppositional measure with documented limitations in criterion validity for elite youth players; therefore, improvements should be interpreted as gains in isolated technical execution rather than direct evidence of enhanced match-level passing performance. Moreover, the absence of match-derived technical and tactical indicators limits conclusions regarding ecological transfer, and the focus on male adolescents over a short intervention period constrains generalization to other populations and long-term adaptations.

Conclusion

In conclusion, the present study suggests that an eight-week SSG–based training program was associated with improvements in aerobic endurance and LSPT performance in adolescent soccer players. The gains observed in Yo-Yo Intermittent Recovery Test Level 1 performance exceeded established thresholds for practical significance, indicating adaptations that may be relevant for match-related physical demands. Improvements in LSPT performance reflect enhanced execution speed and technical efficiency, although these gains should be interpreted as improvements in foundational technical capacity rather than direct evidence of enhanced in-game passing performance. Given the small sample size and the inability to fully disentangle training type from training volume, these conclusions should be interpreted with caution.

Consistent with previous research, these findings support the use of well-structured SSG as an ecologically valid and time-efficient training approach that promotes concurrent physiological and technical development in youth soccer players.

Abbreviations

SSG	Small-sided games
BMI	Body mass index
Yo-Yo IR1	Yo-Yo Intermittent Recovery Test Level 1
LSPT	Loughborough passing test

Supplementary Information

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Additional file 1. CONSORT 2010 checklist (DOCX). Completed CONSORT 2010 checklist of information to include when reporting a randomized trial, with the corresponding section of the manuscript indicated for each item.

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Authors' contributions

SS: Formal Analysis, Investigation, Writing – original draft, Data curation, Methodology, Supervision, Visualization. FCG: Writing – original draft, Writing – review and editing, Investigation, Methodology. GA: Visualization, Methodology, Writing – original draft, Writing – review and editing. FGG: Writing – original draft, Writing – review and editing, Investigation, Methodology, Supervision, Writing – original draft. CS: Visualization, Methodology, Writing – original draft, Writing – review and editing. YEA: Investigation, Methodology, Supervision, Writing – original draft. MFB: Investigation, Methodology, Supervision, Writing – original draft.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to ethical restrictions related to participant confidentiality and the inclusion of sensitive clinical data. However, the data are available from the corresponding authors upon reasonable request, subject to approval by the Health Sciences Ethics Committee of Manisa Celal Bayar University.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the ethical principles of the Declaration of Helsinki. Ethical approval was obtained from the Health Sciences Ethics Committee of Manisa Celal Bayar University (Approval date: July 17, 2025; Decision No: 20.478.4).

Prior to participation, all eligible participants and their legal guardians were provided with detailed verbal and written information regarding the study procedures, potential risks, and benefits. Written informed consent was obtained from all participants and their parents/guardians before inclusion in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

[1]Department of Coaching Education, Faculty of Sport Sciences, Manisa Celal Bayar University, Manisa, Türkiye
[2]Faculty of Sports Sciences, Istanbul Gelisim University, Istanbul, Türkiye
[3]Department of Coaching Education, Faculty of Sport Sciences, Pamukkale University, Denizli, Türkiye
[4]Gulhane Faculty of Physiotherapy and Rehabilitation, University of Health Sciences, Ankara, Türkiye
[5]Faculty of Sports Sciences, Department of Coaching Education, Iğdır University, Iğdır, Türkiye
[6]Faculty of Sports Sciences, Bitlis Eren University, Bitlis, Türkiye

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